

CLASSIFICATION

Global diversity is estimated to range from 10 million species to billions. Nearly 2 million have been described, with many more added each year. All species are named and classified using a system devised over 250 years ago.

For centuries people have studied the natural world. Initially they were limited to what they could find locally and to reports from travellers, as it was impossible to preserve and send specimens any distance. Later, as travel became easier, explorers were paid to collect plants and animals and ship's artists would draw them. By the early 1600s, natural history collections in Europe were substantial, and many specimens had been described, but there was no formal arrangement that made these specimens or accounts of them easily accessible.

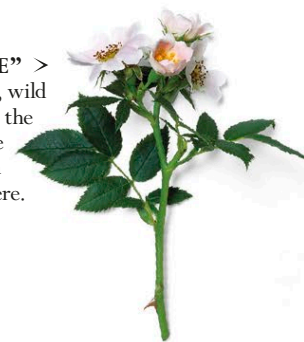
The aim of the early taxonomists, or scientists who describe and classify species, was simple – to organize living things so that they reflected God's plan of creation. Between 1660 and 1713, John Ray published works on plants, insects, birds, fish, and mammals, forming his groups on the basis of morphological (structural) similarities.

The science of morphology, along with other criteria such as behaviour and modern genetics, forms the basis of classification today. In 1758, the tenth edition of *Systema Naturae* was published. It was written by the Swedish botanist Carolus Linnaeus. He and his friend Peter Artedi had decided to divide the natural world between them and classify everything in it, fitting the 7,300 described species into the same hierarchical framework. Although Artedi died before his book was finished, Linnaeus completed the work and published it along with his own.

LATIN NAMES

All living things now have a unique Latin name – such as *Panthera leo* for the lion – which is made up of the genus name, starting with a capital letter, and a descriptive species name. Linnaeus devised this binomial method

“DOG ROSE” >
Known variously as the dog rose, wild briar, witches' briar, dogberry, and the hip tree, this plant has only one Latin name, *Rosa canina*, which identifies it for everyone everywhere.



< “PUMA”

The puma is also known as the cougar or mountain lion. Its Latin name, *Puma concolor*, alludes to its uniform colour.

for identifying different organisms to replace the arbitrary descriptions that existed previously. The new method put an end to confusion caused by the same name being given to several species, or to single species being known by several names.

Within a species, distinct subspecies can sometimes be recognized in different locations. In the 1800s, Elliot Coues and Walter Rothschild adopted a trinomial Latin system to accommodate them. This convention for naming species and subspecies is still used today.

TRADITIONAL CLASSIFICATION



DOMAIN Eukaryota

The latest taxonomic unit to be created is the domain. It is based on whether organisms possess cells with a nucleus (Eukaryota: protists, plants, fungi, and animals) or not (Archaea and Bacteria).



KINGDOM Animalia

In recent years, the traditional kingdoms of plants and animals have been further subdivided. Kingdom Animalia now includes only multicellular organisms that must eat other species in order to survive.



PHYLUM Chordata

A phylum (known as a division in plants) is a major subdivision of a kingdom and is made up of one or more classes that share certain features. Members of the phylum Chordata have a notochord – the precursor of the backbone.



CLASS Mammalia

Introduced by Carl Linnaeus, a class contains one or more orders. Class Mammalia includes only animals that are warm-blooded, have fur, a single jaw bone, and that suckle their young on milk.

> CONTRIBUTING TO CLASSIFICATION

Over the years many scientists have attempted to organize the natural world, combining earlier ideas with new research, culminating in the traditional system of classification shown above and the use of bi- and trinomial Latin names. Some scientists were particularly influential, making significant contributions to taxonomy.

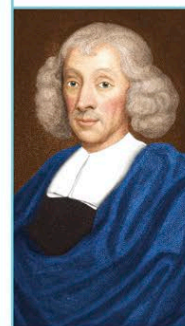
ANIMALS AND PLANTS



Aristotle was the first person to classify living things, and he introduced the term *genos* (meaning race, stock, kin) – genus in Latin. He separated animals into those with blood and those without, not realizing blood need not be red. This division is very close to the modern classification of vertebrates and invertebrates.

ARISTOTLE, 384–322 BCE

ORDER FROM CHAOS



John Ray classified organisms based on their overall morphology rather than just a part of it. In doing so he could establish relationships between species more easily and organize them into groups more effectively. He also divided the flowering plants into two major groups of orders – monocotyledons and dicotyledons.

JOHN RAY, 1627–1705